

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (All Branches) Examination
May 2011

CL103 Mechanics of Solids

Date: 18.05.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Find center of gravity of an I section having top flange 240mm x 10mm, bottom flange 180mm x 10mm and web 300mm x 10mm. 07

Q - 2 (a) Define Limiting Friction, Angle of Friction, Angle of Repose and Coefficient of Friction. What are laws of static friction? 07

(b) Find the forces in all members of the truss shown in Fig. 1. 07

OR

Q - 2 (a) A ladder 6m long rests on horizontal ground and leans against a smooth vertical wall at an angle of 20° with vertical. Its weight is 900 N acting at its middle. It is on the point of sliding when a man weighing 500 N stands on it at a distance 2m along the ladder from foot of the ladder. Calculate the coefficient of friction. 07

(b) Find the forces in all members of the truss shown in Fig. 2. 07

Q - 3 (a) State & prove law of parallelogram. 04

(b) Find resultant of the force system shown in Fig. 3. 05

(c) Find resultant of the force system shown in Fig. 4 with respect to point A. 05

OR

Q - 3 (a) Explain principles of superposition & transmissibility. 04

(b) The sum of two concurrent forces P and Q is 270 N and their resultant is 180 N. The angle between the force P and resultant R is 90° . Find the magnitude of each force and angle between them. 05

(c) What is Moment? Differentiate between moment and couple. Give characteristics of a couple. 05

SECTION - II

Q - 4 (a) Attempt any one: 03

(i) Define Amplitude, Frequency and Natural time period.

(ii) Derive equation of un-damped SDOF system.

(b) Describe how you will measure the hardness of the given metal in the laboratory. 04

Q - 5 (a) A stepped bar is loaded as shown in Fig. 5. Calculate the stresses in each part and total change in the length of the bar. Take $E_s = 200$ GPa, $E_c = 100$ GPa and $E_b = 80$ GPa. 07

OR

Q - 5 (a) A load of 2000 kN is applied on a short concrete column 500mm x 500mm, 07

reinforced with four 10mm steel bars. Find stresses in concrete and steel. Take value of E for steel as $2.1 \times 10^5 \text{ N/mm}^2$ and for concrete $1.4 \times 10^5 \text{ N/mm}^2$.

Q - 5 (b) Define Poisson's ratio, Bulk modulus

02

Q - 5 (c) A bar 0.5m long is rectangular in section with width 40mm and thickness 30mm. Calculate the change in dimensions when a tensile load of 120 kN is acting along its longitudinal axis if $E = 200 \text{ kN/mm}^2$ and Poisson's ratio 0.25.

05

OR

Q - 5 (c) A steel rod of 16mm diameter and 5m length, is connected to two grips at each end at a temperature of 150°C . Find the pull exerted when the temperature falls to 40°C if the ends (i) Do not yield (ii) Yield by 1 mm.

05

Take coefficient of thermal expansion as $12 \times 10^{-6} \text{ per } ^\circ \text{C}$.

Q - 6 Draw Shear Force & Bending Moment diagrams for any two beams from figures Fig. 6(a), Fig. 6(b) and Fig. 6(c) indicating values at all important points. Find point of contra-flexure if any.

14

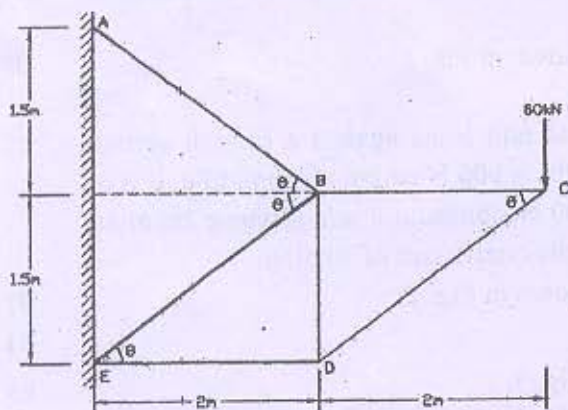


Fig. 1

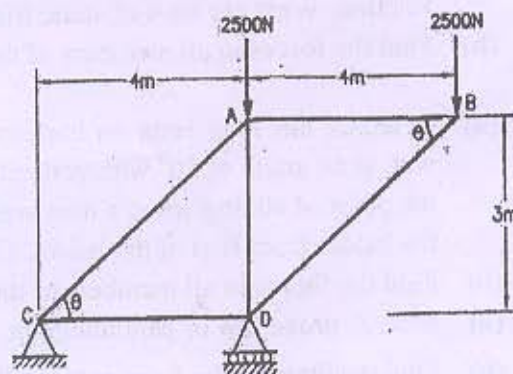


Fig. 2

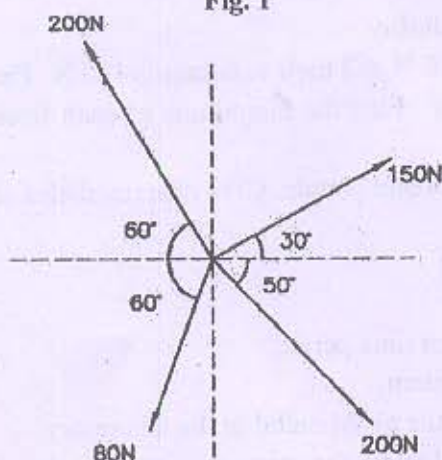


Fig. 3

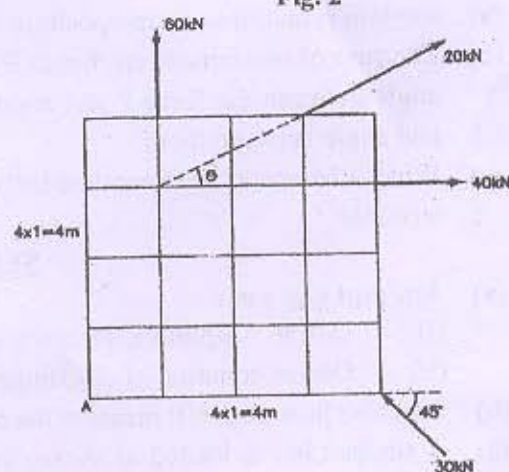


Fig. 4

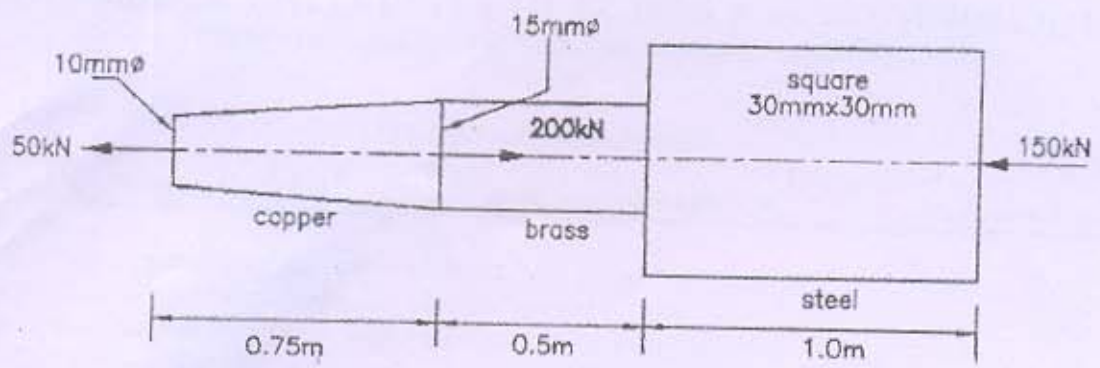


Fig. 5

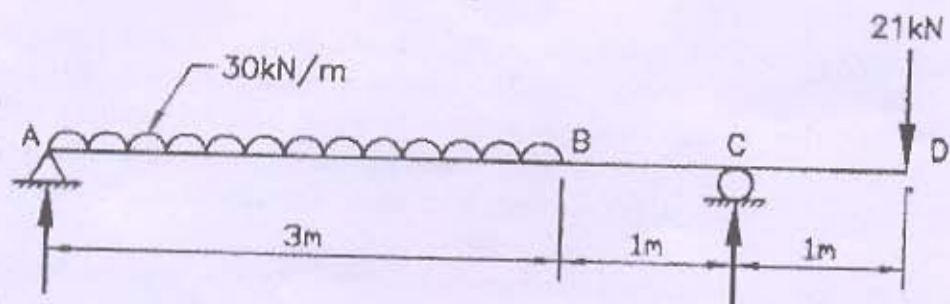


Fig. 6(a)

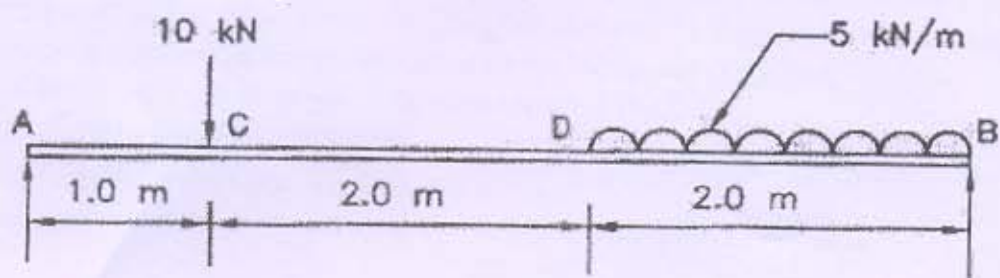


Fig. 6(b)

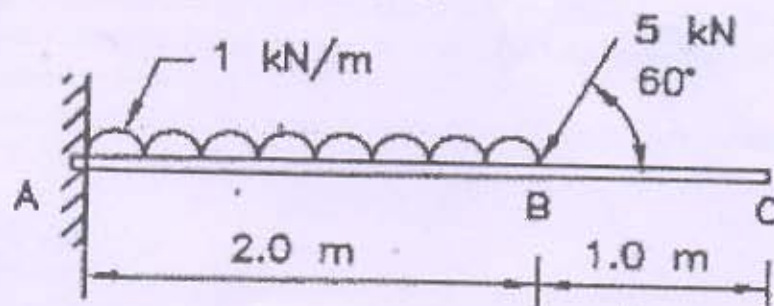


Fig. 6(c)
